Sybil-Resistant Federated Learning via Blockchain-Based Economic Accountability

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***Abstract*—Federated Learning (FL) in trustless, decentralized environments faces three critical vulnerabilities: susceptibility to Sybil attacks, model poisoning from adversarial participants. In addition, a widespread reliance on global validation sets that compromise data privacy. Likewise, the paper proposes a two-layer blockchain framework coupled with an EMH-DNN to address these challenges. Layer 1 implements Sybil-resistant identity governance through PoP registration and a decentralized median-voting PoC protocol. In addition, it eliminates the need for a shared global validation dataset. Similarly, Layer 2 monitoring relies solely on each committee member's private local test partition, with only aggregated metrics committed on-chain. Layer 2 introduces economic accountability via a staking mechanism and a blockchain-based state-rollback procedure that enables retroactive correction of poisoned model updates. The EMH-DNN architecture employs parallel feature-extraction branches with complementary dropout and client-specific output heads for shard-routing prediction under non-IID data distributions. Experimental evaluation on an approximate nearest neighbor routing task over the SIFT1B benchmark demonstrates that the framework maintains ∼93.8% Recall@1000 under a 50% Sybil attack across configurations of $N \in \{4, 8\}$ clients, while reducing the Attack Success Rate (ASR) from 100% to 0.0% through retroactive state correction.**

***Index Terms*—Federated Learning, Blockchain, Sybil Attack, Model Poisoning Defense, Proof-of-Personhood, State Rollback**

I. Introduction

FL enables distributed clients to collaboratively train a shared model without exchanging raw data [1], [2]. However, the standard FL depends on a central server for aggregation – a single point of failure, trust, and control. *Blockchain-based FL* replaces this server with a decentralized ledger: clients train locally and submit model updates as blockchain transactions, where a consensus mechanism validates and aggregates contributions into a new global model. Similarly, it provides immutability (updates cannot be tampered with), auditability (every update is recorded), and decentralization (no single entity controls aggregation) [9], [10]–properties essential for privacy-sensitive applications where participants may not trust one another.

Evolution and limitations of the field. Early blockchain-FL systems such as BlockFL [3] and FLChain [4] demonstrated feasibility but employed flat architectures with no economic accountability or Sybil resistance. Similarly, Committee-based designs followed: BFLC [1] introduced cross-validation committees, while Biscotti [5] combined Proof-of-Federation with Multi-Krum, tolerating up to 48% adversaries. VeryFL [6] added verifiable smart contracts, and VFChain [7] and VerifBFL [8] explored verifiable computation via authenticated data structures and zk-SNARKs. Reliance on shared global validation data to score contributions undermines the fundamental privacy FL aims to provide.

Decentralized consensus mechanisms are fundamentally threatened by the Sybil attack. Specifically, a vulnerability in which a lone adversary generates numerous fictitious identities. Douceur [11] proved that Sybil resistance is impossible without some form of identity verification, motivating PoP protocols [12]–[14]. Within FL, FoolsGold [15] and FedSybil [16] detect Sybils via gradient similarity, but all such defenses are *reactive*—they attempt post-entry detection that sophisticated adversaries can evade [17].

On the aggregation front, robust rules such as Krum [18], coordinate-wise Median [19], FLTrust [20], and FLAME [21] offer statistical guarantees, but *catastrophically fail under non-IID data* [23], [24], and optimized attacks [17], [26] consistently circumvent them. Likewise, an inherent arms race that statistical defenses alone cannot resolve [25].

Economic mechanisms have been explored in blockchain consensus (Casper FFG's accountable safety [27]; Polkadot's slashing [28]) and FL incentives (DeepChain [29]; Kang et al.'s contract theory [30]; BlockFLA [31]), but none include *model rollback*. Specifically, they penalize financially without correcting the poisoned model. Conversely, model recovery methods (FedRecover [32], Crab [33]) restore clean states but operate centrally without financial deterrence. Game-theoretic defenses (FedGame [34]) model adversarial dynamics but carry no real economic consequences.

Three open problems. Across this landscape, three critical gaps persist: **(P1)** contribution evaluation requires global validation sets, violating privacy; **(P2)** committed poisoned updates are irreversible on blockchain; **(P3)** validators face no economic consequence for approving malicious updates.

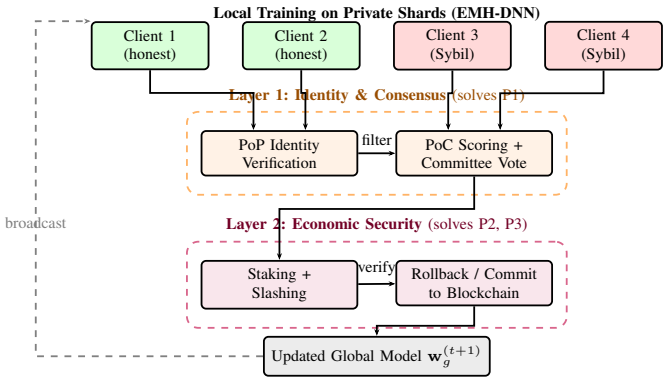


Fig. 1. System workflow. Clients train locally using EMH-DNN and submit updates. Layer 1 filters Sybil identities via PoP and evaluates quality through committee-based PoC voting on private data (P1). Layer 2 enforces staking and triggers rollback upon detecting poisoning (P2, P3). The verified global model is broadcast for the next round.

Our approach. This paper addresses all three problems through a two-layer architecture (Fig. 1). **Layer 1** solves **P1** by replacing global validation sets with a PoP identity gate and peer-to-peer median-voting consensus, where committee members evaluate updates using only their *private local data*. **Layer 2** solves **P2** and **P3** by introducing economic staking—validators must lock collateral—and a blockchain-based rollback mechanism that reverts the ledger state upon detecting poisoning, making attacks both *economically irrational* (stake is slashed) and *technically reversible* (model state is restored). To handle non-IID data heterogeneity, the study proposes the EMH-DNN, a parallel-branch architecture with client-specific output heads that mitigates catastrophic forgetting at the routing layer during aggregation.

The principal contributions are: (1) a *two-layer trust architecture* integrating Sybil-resistant identity governance (Layer 1, addressing P1) with economic rollback security (Layer 2, addressing P2–P3); (2) a *privacy-preserving validation protocol* where committee members use only private local data, eliminating global validation sets; and (3) a *slashing-based accountability mechanism* with state rollback enabling retroactive correction of poisoned updates—a capability absent from all prior blockchain-FL frameworks [22], [35]–[37].

II. Methodology

The research considers a decentralized network of N clients that jointly minimize a global loss function $F(\mathbf{w})$ over their respective private datasets $\mathcal{D}_k$, $k = 1, \ldots, N$. The framework operates across two complementary security layers—Layer 1 for identity and consensus, Layer 2 for economic accountability—supported by the EMH-DNN architecture designed for non-IID robustness (Table I).

A. Layer 1: Identity Governance and Decentralized Consensus

Layer 1 provides Sybil resistance and quality-aware model validation without requiring a global validation dataset.

TABLE I
Key Symbols and Parameters

Symbol	Definition	Symbol	Definition
N	Total clients	S_{PoC}	PoC score
$\tau_k^{(t)}$	Trust score of k at t	S_{min}	Min. stake
$\mathcal{C}^{(t)}$	Committee at round t	$P_{penalty}$	Slashing penalty
$\mathbf{w}_k$	Local weights of k	δ_{drop}	Acc. drop threshold
$\mathbf{w}_g^{(t)}$	Global model at t	δ_{th}	PoC acceptance thr.
$Stake_k$	Locked stake of k	$\mathbf{z}$	Fused feature repr.
r_{reward}	Trust reward incr.	$r_{penalty}$	Trust penalty decr.
τ_{min}	Full participation thr.	H_k	Client-specific head
$\mathbf{W}_1, \mathbf{W}_2$	Branch weight matrices	B_1, B_2	Parallel branches

1) Trust Dynamics and Probation: To prevent Sybil clients from immediately gaining full influence, every client k maintains a dynamic trust score $\tau_k^{(t)} \in [0, 1]$, initialized at a low value $\tau_k^{(0)}$ through a Proof-of-Personhood (PoP) registration challenge. New clients enter a *probationary state* below the full participation threshold τ_{min}, ensuring they must demonstrate honest behavior before their updates carry significant weight. Trust evolves asymmetrically after each round:

$$\tau_k^{(t+1)} = \begin{cases} \min(1,\ \tau_k^{(t)} + r_{reward}) & \text{accepted \& honest} \\ \max(0,\ \tau_k^{(t)} - r_{penalty}) & \text{accepted \& malicious} \\ \tau_k^{(t)} & \text{rejected} \end{cases} \tag{1}$$

The asymmetry is deliberate: the reward increment r_{reward} is small, so that reputation is earned gradually over many rounds, while the penalty $r_{penalty}$ is an order of magnitude larger, so that a single detected malicious act causes severe trust loss. This design follows the principle from PoS blockchain economics [27] that the cost of misbehavior must far exceed the cost of honest participation. Probationary clients receive a slightly higher reward rate to enable bootstrapping without granting premature influence. Clients below the minimum trust floor are excluded entirely, while those in probation ($\tau_k < \tau_{min}$) receive reduced aggregation weight (see Eq. 7). Concrete parameter values are reported in Section III-A2.

2) Committee Election and Peer Validation: A validation committee $\mathcal{C}^{(t)}$ is elected each round via score-based selection, prioritizing clients with strong historical validation performance. Committee members must be PoP-registered and, when staking is enabled, hold sufficient stake. Members rotate each round to prevent concentration of validation power.

Because no global validation set exists, the committee evaluates each candidate update $\mathbf{w}_k$ using members' *private local datasets*—the key mechanism that solves problem P1. To quantify update quality without shared data. The study introduces the PoC score. Likewise, which captures three complementary aspects of update utility:

$$S_{PoC}(k) = \gamma_1 \cdot u_{acc} + \gamma_2 \cdot u_{loss} + \gamma_3 \cdot u_{rob} \tag{2}$$

where u_{acc} measures the improvement in accuracy (how much the local update improves on the current global model), u_{loss} captures the reduction in normalized losses (ensuring a scale-invariant comparison between clients), and $u_{rob} = (1 + \|\Delta \mathbf{w}\|/100)^{-1}$ penalizes gradient explosions – a known signature of poison attacks [17]. The coefficients $\gamma_1, \gamma_2, \gamma_3$ control the relative importance of accuracy, loss, and robustness.

An update must exceed a minimum PoC threshold δ_{th} and receive $\geq 2/3$ committee approval. The final contribution weight combines individual quality with the committee's aggregate median assessment:

$$S_{combined} = 0.6 \cdot S_{PoC} + 0.4 \cdot S_{median} \tag{3}$$

Using the median (rather than the mean) ensures robustness to outlier evaluations from any compromised committee members, following the statistical guarantees of coordinate-wise median aggregation [19].

B. Layer 2: Economic Security and Rollback

Critically, no global dataset is introduced: each committee member independently evaluates the updated model on its own private local test partition, and the median Recall@1000 is computed on-chain — no raw data leaves any client. Layer 1's peer validation filters overtly malicious updates, but sophisticated poisoning attacks may pass initial quality checks yet degrade model performance over subsequent rounds. Layer 2 addresses this by making such attacks *economically irrational* (solving P3) and *technically reversible* (solving P2).

1) Staking and Slashing: The key insight, borrowed from PoS blockchain consensus [27], is that validators who have financial skin-in-the-game are incentivized to verify honestly. Committee eligibility therefore requires locking a minimum stake $Stake_k \geq S_{min}$. When committee $\mathcal{C}^{(t)}$ approves a block B_t, members' stakes remain locked until the block's impact is verified. If global model quality Acc_t drops below a safety margin δ_{drop} relative to the previous round, a *slashing event* penalizes the responsible validators:

$$Stake_j \leftarrow \max(0,\ Stake_j - P_{penalty}), \quad \forall j \in \mathcal{C}^{(t)} \tag{4}$$

Simultaneously, trust scores are reset to $\tau_j \leftarrow 0$, permanently banning the validators. The penalty $P_{penalty}$ is calibrated relative to S_{min} so that a single slashing event eliminates a substantial fraction of stake, making the expected cost of approving a poisoned update exceed any potential gain from collusion.

2) Blockchain Rollback: Upon detecting poisoning, the rollback mechanism executes a three-step state reversion: (1) invalidate block B_t, (2) revert the global model to the last verified checkpoint $\mathbf{w}_g^{(t-1)}$, and (3) resume training with a freshly elected committee. This capability directly addresses P2: unlike standard blockchain-FL where immutability makes corruption permanent, our "time machine" mechanism ensures that no single poisoned round can permanently compromise the global model.

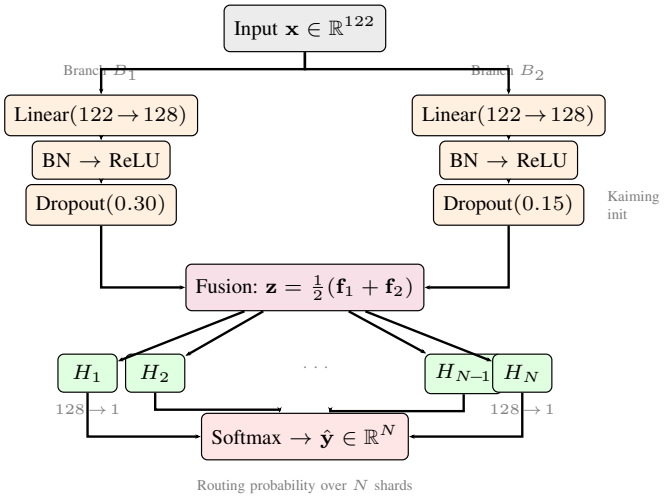


Fig. 2. EMH-DNN architecture. Two parallel branches with complementary dropout extract diverse features, fused into a shared representation $\mathbf{z}$. Per-client heads H_k produce scalar logits, normalized via softmax into a routing probability over N shards.

C. Model Architecture: Enhanced Multi-Head DNN

To mitigate catastrophic forgetting under non-IID data distributions. The research introduces the EMH-DNN, which decouples shared feature extraction from client-specific routing prediction, isolating shard-level routing knowledge in dedicated output heads while relying on trust-weighted aggregation (Eq. 7) to stabilize the shared representation. Figure 2 illustrates the complete architecture.

1) Parallel Feature Extraction: The network core consists of two parallel branches B_1 and B_2 processing $\mathbf{x} \in \mathbb{R}^d$ simultaneously (Fig. 2). Let $\mathbf{W}_1, \mathbf{W}_2 \in \mathbb{R}^{h \times d}$ denote their weight matrices ($h = 128$). *Why two branches?* In non-IID federated settings, a single feature extractor risks overfitting to the local distribution of whichever client dominates aggregation. Dual branches with *complementary dropout rates* address this: B_1 applies higher dropout (rate p), which acts as an implicit ensemble and promotes generalized, distributed representations [24]; B_2 uses lower dropout ($0.5p$), preserving fine-grained features that B_1 may discard. Their fusion combines robustness with fidelity:

$$\mathbf{z} = \frac{1}{2}\big(\phi(\mathbf{W}_1\mathbf{x}) + \phi(\mathbf{W}_2\mathbf{x})\big) \tag{5}$$

where $\phi(\cdot)$ denotes the pipeline of linear transformation, batch normalization, ReLU, and branch-specific dropout. Weights are initialized via Kaiming normal. The specific dropout rates are reported in Section III-A2.

2) Client-Specific Multi-Head Output: The shared representation $\mathbf{z}$ feeds into N distinct output heads $\{H_1, \ldots, H_N\}$, one per client shard. Each head $H_k : \mathbb{R}^h \to \mathbb{R}$ produces a scalar logit, and the concatenated outputs are normalized via softmax to yield a routing probability distribution over shards:

$$\mathbf{y} = \text{Softmax}\big[H_1(\mathbf{z}),\ H_2(\mathbf{z}),\ \ldots,\ H_N(\mathbf{z})\big]^\top \tag{6}$$

where $\mathbf{y}[j]$ represents the predicted fraction of relevant data residing on shard j.

3) Trust-Weighted Aggregation: To limit Sybil influence during model aggregation, base weights α_k (proportional to data size) are scaled by a nonlinear trust-dependent factor. Because trust scores satisfy $\tau_k \in [0, 1]$, raising them to a power $p > 1$ amplifies the gap between high- and low-trust clients. The study use cubic scaling for probationary clients and quadratic for trusted ones:

$$\alpha_k^{(trust)} = \alpha_k \cdot \begin{cases} \tau_k^3 & \tau_k < \tau_{min} \text{ (probationary)} \\ \tau_k^2 & \tau_k \geq \tau_{min} \text{ (trusted)} \end{cases} \quad (7)$$

The stronger cubic suppression for probationary clients ensures that newcomers—who have not yet demonstrated sustained honest behavior—contribute minimally to aggregation even if they pass PoC screening, following the principle that initial trust must be earned, not assumed [20].

4) Deployment in the Federated Setting: The *server* instantiates a single global EMH-DNN with N output heads matching the number of client shards. Each round, eligible client k receives a deep copy of $\mathbf{w}_g^{(t)}$ and trains locally on its private SIFT shard $\mathcal{D}_k$ for 3 epochs. Because the training labels are normalized routing distributions over *all* N shards (Section III-A1), client k's loss function produces gradients for the shared branches (B_1, B_2) and all N heads simultaneously: the ground-truth vector $\mathbf{y}[i]$ assigns nonzero probability to every shard containing at least one of the query's top-1000 neighbors, so the MSE loss naturally propagates learning signal to each head H_j in proportion to shard j's true neighbor density. In practice, the dominant gradient flows to client k's own head (which holds the largest neighbor share), while the remaining heads receive weaker but nonzero supervision. Each head is a single linear layer ($128 \to 1$), so the N-head overhead is $\mathcal{O}(Nh)$ parameters—negligible relative to the shared branches when N is small (4–8 in our experiments); for larger federations, head pruning or sparse communication could further reduce this cost.

This architectural separation ensures: (i) the shared encoder converges to a universal embedding across shards, and (ii) the per-head isolation *mitigates* catastrophic forgetting *at the routing layer*, since each shard's routing knowledge is maintained in a dedicated head. Forgetting in the shared branches due to non-IID updates remains possible and is addressed by trust-weighted aggregation (Eq. 7), which limits low-trust clients' influence on the shared representation. Validated updates are committed to the blockchain. To trigger Layer 2 rollback, each committee member independently evaluates the updated global model on its own local test partition; if the median Recall@1000 across committee members drops below the safety margin δ_{drop}, the slashing and rollback protocol (Section 2.2) is activated.

D. Protocol Summary

Algorithm 1 summarizes the complete protocol.

Algorithm 1 Secure Two-Layer Federated Learning

Initialize: Model $\mathbf{w}_0$, trust $\tau_k^{(0)} = 0.3\ \forall k$, stakes S
for round $t = 1, \ldots, T$ **do**
  // Layer 1: Consensus
  Elect committee $\mathcal{C}^{(t)}$ via score-based selection
  $\mathcal{P}_{valid} \leftarrow \emptyset$
  for each eligible client k **in parallel do**
    $\mathbf{w}_k \leftarrow \text{LocalTrain}(\mathbf{w}_g^{(t-1)}, \mathcal{D}_k)$ using EMH-DNN
    Compute $S_{PoC}(k)$ per Eq. (2)
    if $S_{PoC}(k) \geq \delta_{th}$ **then**
      Committee validates; accept if $\geq 2/3$ approve
      if accepted **then**
        $\mathcal{P}_{valid} \leftarrow \mathcal{P}_{valid} \cup \{(\mathbf{w}_k, S_{combined}(k))\}$
      end if
    end if
  end for
  // Trust-weighted aggregation:
  $\mathbf{w}_g^{(t+1)} \leftarrow \sum_{k \in \mathcal{P}_{valid}} \alpha_k^{(trust)} \cdot \mathbf{w}_k$
  Mine block B_t with validated updates
  // Layer 2: Economic Security
  Each committee member evaluates Acc_t on its private local test partition; compute median on-chain.
  if $\text{Acc}_t < \text{Acc}_{t-1} - \delta_{drop}$ **then**
    Rollback to $\mathbf{w}_g^{(t-1)}$; slash $\mathcal{C}^{(t)}$; set $\tau_j \leftarrow 0$
  else
    Update trust per Eq. (1); commit B_t
  end if
end for

III. Experimental Evaluation

A. Experimental Setup

1) Task, Dataset, and Evaluation Metric: The study evaluate the framework on an *approximate nearest neighbor (ANN) routing* task—a core operation in large-scale retrieval systems where a database is sharded across distributed clients.

Dataset. The research use the SIFT1B benchmark [38], a standard billion-scale dataset of Scale-Invariant Feature Transform (SIFT) descriptors widely used for evaluating ANN search. SIFT descriptors are 128-dimensional gradient-histogram vectors extracted from image keypoints; our pre-processed subset uses 122-dimensional vectors. The dataset is partitioned across N client shards with high data skew (non-uniform shard sizes and distributional heterogeneity) to simulate realistic non-IID federated scenarios. Two configurations are tested: $N \in \{4, 8\}$, with 50% of clients designated as Sybil attackers.

Learning task. Given a query vector $\mathbf{x} \in \mathbb{R}^{122}$, the true top-1000 nearest neighbors are distributed across the N shards. The ground-truth label $\mathbf{y}^{raw}[i] \in \mathbb{Z}^N$ records the count of true top-1000 neighbors on each shard (summing to 1000), and is row-normalized to a probability distribution $\mathbf{y}[i] = \mathbf{y}^{raw}[i]/1000$. The EMH-DNN learns to predict the routing distribution $\hat{\mathbf{y}}[i]$ by minimizing MSE loss against $\mathbf{y}[i]$, so that queries can be routed to the shard(s) most likely to contain relevant neighbors.

Recall@1000. Following standard retrieval evaluation, Recall@1000 is the fraction of true top-1000 neighbors found in probed shards: $\text{Recall@1000} = \sum_{j \in \mathcal{S}_K} \mathbf{y}^{raw}[i][j] \,/\, 1000$, where $\mathcal{S}_K$ contains the K shards with highest predicted probability. The study report top-1 routing ($K = 1$): the query is sent to $\arg\max_j \hat{\mathbf{y}}[j]$ only, and recall measures how many true neighbors reside on that shard.

TABLE II
PERFORMANCE AND SECURITY METRICS ON SIFT DATASET (20 ROUNDS, 50% SYBIL ATTACK)

Metric	$N = 4$	$N = 8$
Model Utility		
Final Test Recall@1000	93.93%	93.72%
Final Test Loss (MSE)	0.0018	0.0008
Security & Trust		
Honest Trust (τ_{honest})	0.41	0.39
Sybil Trust (τ_{sybil})	0.15	0.15
Peak ASR (Round 1)	100.0%	100.0%
Final ASR (Round 20)	**0.0%**	**0.0%**
Total Sybil Updates Accepted	2	4

2) Model and Protocol Configuration: The EMH-DNN (Fig. 2) is trained with MSE loss using SGD (lr $= 0.01$, momentum $= 0.9$) for 3 local epochs per round with batch size 128. Committees comprise 30% of eligible clients. Trust: $\tau_k^{(0)} = 0.3$, $\tau_{min} = 0.45$, $r_{reward} = 0.02$ (0.05 probationary), $r_{penalty} = 0.15$. PoC: $\gamma_1 = 0.4$, $\gamma_2 = 0.4$, $\gamma_3 = 0.2$; δ_{th} adapts from 0.02 to 0.1. Layer 2: $\delta_{drop} = 0.05$, $S_{min} = 10.0$, $P_{penalty} = 5.0$.

3) Attack Model: Adversarial clients execute a coordinated 50% Sybil attack using *sign-flip* poisoning (negating gradient directions to maximally degrade the global model) combined with free-riding (zero/minimal updates), while attempting to maintain trust scores above the participation threshold.

B. Performance Analysis

1) Global Model Convergence: Despite heavy adversarial presence, the system demonstrated robust convergence over 20 communication rounds (Table II). The global model achieved Recall@1000 of **93.93%** for $N = 4$ and **93.72%** for $N = 8$ under top-1 shard routing. MSE test loss decreased consistently to 1.81×10^{-3} ($N = 4$) and 8.12×10^{-4} ($N = 8$), confirming that the decentralized committee validation and trust-weighted aggregation successfully filtered sign-flip poisoned updates.

2) Sybil Resistance and Trust Dynamics: In Round 1, the system operated in a "trust-but-verify" mode, resulting in a 100% Attack Success Rate as initial Sybil updates were tentatively accepted (2 updates for $N = 4$, 4 updates for $N = 8$). However, Layer 2 mechanisms responded immediately: by Round 2, the ASR dropped to **0.0%** and remained there for all subsequent rounds. Honest clients stabilized at $\tau \approx 0.41$ ($N = 4$) and $\tau \approx 0.39$ ($N = 8$), while Sybil clients were penalized from $\tau = 0.3$ to the minimum floor of $\tau = 0.15$, effectively locking them out of committee participation. The trust dynamics demonstrate the effectiveness of combining PoP-based identity verification with economic penalties.

3) Committee Fairness: Honest clients served as committee members in 4–5 rounds ($N = 4$) and 2–4 rounds ($N = 8$), demonstrating balanced validation responsibilities. Sybil clients were excluded after initial penalization, confirming the trust-weighted election prevents adversarial concentration of power.

TABLE III
COMPARISON WITH STATE-OF-THE-ART SECURITY STANDARDS

Metric	**Industry** [36], [37]	**Research** [22], [23], [35]	**Ours**
Security Resilience			
ASR	$< 10\%$	$< 1\%$	**0.0%**
Attack Tolerance	20%–33%	33%–45%	**50%**
Sybil Handling	Filtering	Statistical	**Rej. + Banned**
Model Utility			
Accuracy Drop	$< 5\%$	$< 3\%$	$\approx$ **0.0%**
Final Recall@1000	$> 95\%$	$> 98\%$	**93.93%**
Comp. Overhead	High	Moderate	**Low**

C. Comparison with State-of-the-Art

Table III benchmarks the proposed framework against established standards, including VeryFL's verifiable aggregation [6], FL-Defender's gradient-based defenses [22], and applications in critical infrastructure domains [35]–[37].

The proposed framework achieves **0.0%** ASR, surpassing both industry ($< 10\%$) and research ($< 1\%$) benchmarks, with **50%** attack tolerance exceeding the 33%–45% threshold typical of Byzantine-robust methods [23]. Unlike statistical filtering approaches, Layer 2 *rejects and permanently bans* malicious clients through economic slashing and state rollback, while maintaining low computational overhead due to the lightweight median-voting consensus.

IV. DISCUSSION AND LIMITATIONS

The two-layer design provides complementary security. For instance, Layer 1 offers proactive defense through trust-gated participation and peer validation. Likewise, Layer 2 detects and reverses poisoning that evades initial screening. Subsequently, replacing global validation sets with peer-to-peer evaluation on private local data, the framework preserves FL's privacy guaranty an improvement over systems such as BlockFL [3]. The staking mechanism ($P_{penalty} = 5.0$ against $S_{min} = 10.0$) eliminates half a validator's stake per slashing event, with trust reset to zero permanently excluding them, making sustained collusion prohibitively expensive.

Several limitations warrant acknowledgment. The framework has been evaluated only on small networks ($N \leq 8$); large-scale deployments may require hierarchical committee architectures and techniques to manage the linearly growing N-head communication cost (e.g., head pruning, sparse updates, or federated distillation). Similarly, the multi-head design mitigates catastrophic forgetting at the routing layer but does not fully prevent representation drift in the shared branches under extreme non-IID heterogeneity; complementary techniques such as FedProx regularization could strengthen this. The system assumes PoP-based trust initialization, which may be challenging in fully anonymous environments. The staking requirement creates entry barriers for resource-constrained participants, and the peer validation process introduces communication overhead compared to centralized approaches. Future work includes adaptive trust initialization for trustless bootstrapping, differential privacy guarantees for peer validation, tiered incentive mechanisms balanc-

ing security with accessibility, and extending the framework beyond EMH-DNN. All intensive operations — training, PoC scoring, and evaluation — execute off-chain; only contribution scores, model hashes, and stake transactions are committed on-chain, keeping overhead minimal while preserving privacy.

V. Conclusion

The paper presented a two-layer blockchain framework for secure decentralized FL. Likewise, Layer 1 eliminates reliance on global validation sets through PoP identity governance and decentralized median-voting consensus. Subsequently, Layer 2 introduces economic accountability via staking, slashing, and blockchain-based rollback. More precisely, that renders poisoning both economically irrational and technically reversible. The EMH-DNN architecture supports non-IID robustness through parallel feature extraction and client-specific output heads.

Experimental evaluation on an ANN routing task over the SIFT1B benchmark confirms $\sim$93.8% Recall@1000 under a 50% Sybil attack while reducing ASR from 100% to 0.0% within a single round of retroactive correction. These results establish that privacy-preserving, economically accountable federated learning is achievable in adversarial decentralized environments.

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